HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained a **Python-based platform** for quantitative researchers, enabling efficient interaction with trading systems and facilitating rapid data analysis.
- Designed and implemented **production-quality data pipelines** to process and manage large datasets for algorithmic trading strategies, ensuring data integrity and accessibility.
- Collaborated with **Traders and Quantitative Researchers** to translate new trading behaviors into software requirements and implement corresponding platform features.
- Enhanced system stability and performance for the quantitative study environment, leading to a **15% reduction in query execution times** for complex analytical tasks.
- Participated in the full software development lifecycle, including code reviews, testing, and deployment of new features in an **agile environment**.
- Containerized key platform services using **Docker** and managed deployments on **AWS EC2 instances**, improving environment consistency and simplifying rollout procedures.
- Automated routine data validation and reporting tasks using **Python scripts**, reducing manual effort by approximately **8 hours per week** for the operations team.

SKILLS

Languages – Python, Java, SQL, R, C#

Frameworks & Libraries – Spring Boot, pandas, NumPy, scikit-learn, FastAPI, Flask, Hibernate, JPA

Databases – PostgreSQL, MySQL, Oracle, NoSQL

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Jenkins, Git

Tools & Methodologies – Jira, Confluence, Agile, Scrum, Linux, Splunk, Dynatrace, Power BI, Tableau

PROJECTS

Python Trading Strategy Environment

- Developed a core **Python platform** for quantitative researchers to analyze market data and backtest trading strategies, supporting a team of 5 researchers.
- Integrated with multiple **relational databases (PostgreSQL, Oracle)** to ingest historical market data and store simulation results, managing datasets up to 5TB.
- Implemented automated data quality checks and reporting using **pandas**, identifying and flagging data anomalies that could impact study accuracy.

High-Throughput Data Pipeline for Options Analysis

- Engineered a **Python-based data pipeline** to process real-time options trading data streams, handling approximately **2,500 messages per second** during peak market hours.
- Built RESTful APIs using **FastAPI** to serve processed data to downstream analytical services and research environments, ensuring low-latency access.
- Containerized the pipeline components with **Docker** for deployment on an **AWS EKS cluster**, enhancing scalability and resilience.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)